Lesson 6: Teach a Machine

Module 2 | 60 minutes | Machine learning from data | Grades 6–11 · ages 11–16

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. machine learning ___ A. Let the computer study the examples and learn the pattern.
b. class ___ B. The examples you give the model to learn from.
c. data ___ C. One group the model learns to recognise, like “Paper”.
d. train ___ D. When a computer learns patterns from examples.

2. Prompt Lab: my data diary plan

Write your data-diary plan for your Teachable Machine model. Name your classes, say how many example photos you will collect, how you will make the data varied, how you will test it, and one failure you expect. Write your prompt here, then try it in Teachable Machine.

3. The ML loop

Put the machine-learning steps in order Number them 1–6.

a. ___ Find the mistakes and add better data
b. ___ Test it with new examples
c. ___ Train the model
d. ___ Collect varied examples for each class
e. ___ Choose your classes
f. ___ Train and test again

4. Helps the model, or confuses it?

Write each item in the right group: Helps the model / Confuses the model.

a. Photos from many different angles → _____
b. Every “Paper” photo taken on the same white desk → _____
c. A “Nothing” class with an empty background → _____
d. Putting a few plastic photos in the Paper class by mistake → _____
e. Photos in bright light and dim light → _____
f. Only 3 photos per class → _____

5. Why did it fail?

Zoya’s rock-paper-scissors model works perfectly for her hand but often fails for her brother’s hand. Why? Circle one.

a. A. The model is broken
b. B. It was trained only on Zoya’s hand, so it has not learned other hands
c. C. Her brother is doing it wrong
d. D. Teachable Machine only works for one person

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can explain what machine learning is.
☐ I can collect data, train and test a model in Teachable Machine.
☐ I can find where my model fails and explain why.
☐ I can improve a model with better data.